# Data Premier League 2026

## Preliminary Round

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*Question 2 - 2023 vs 2024 Comparative Analyses*

*All 10 teams and the full 2023 and 2024 IPL seasons*

*30 September 2026*

# Context and Analytical Objective

- **Question Selection and Rationale:** Q2 was chosen since the given data has complete population for both years which allows for well-powered analyses at the league and phase level, i.e. a before and after, which fits a tactical evolution framing nicely.  However it is one comparison and no causality can be inferred.

- **Seasons Chosen and Franchise Lens:** All 10 teams, the full 2023 and 2024 seasons, assessed by the analytics staff of a franchise to read the league.

- **Core Analytical Objective:** To establish empirically if there was a change in play between 2023 and 2024 so that a franchise's staff can decide based on evidence how much batting risk to carry in what phases and innings, and whether a change in batting strategy can be answered with who bowls or what is bowled.  Selection, recruitment and impact player substitution as used is described.

- **Scope and Boundaries of Analyses:** League and phase levels comparisons have good statistical power, team level comparisons less so i.e. can only detect large changes; individual player year-over-year comparisons are thin cells and generally kept out of scope of the analyses.  Data is only two full seasons, so no causal claims are made. Venue and pitch effects, within season dynamics and predictions are also out of scope of the analyses.

# Data and Methodology

## Data Sources

*Provided ball-by-ball file: IPL 2023 and 2024, 145 matches, 34,967 deliveries, about 60 fields*

- Fields: match, player and team context; the delivery's outcome; running match state; hand-tagged line, length, shot and control; tracked speed and release point; shot placement; provided win-probability and projected-score model outputs.
- *Cricsheet* match records IPL 2023 and 2024 (cached, checksummed): impact-substitute events and the twelve named per team-match.
- *Provided auction tables, both seasons*: squad status, type and price per player, joined to the ball-by-ball by reconciled names.

## Cleaning and Prep

*Every check passed or was explained; no rows were removed; every fix is a new column beside the raw one*

- Runs, balls and wickets reconciled ball by ball against the running totals; two allowances of zero repaired from balls bowled plus remaining.
- Conventions recovered and corrected: economy from a laws-correct bowler charge (no byes or leg byes); wickets from the dismissal flag, not the dismissed-player column, which is filled on every ball; control nulled on wides where it had leaked.
- Zero distinguished from missing per column: (0, 0) shot placement is a null placeholder, not a position.
- Hand-tagged columns carry 7% missingness at max, control missingness is same in both seasons: cancels in a season difference, widens the true interval, marked wherever used

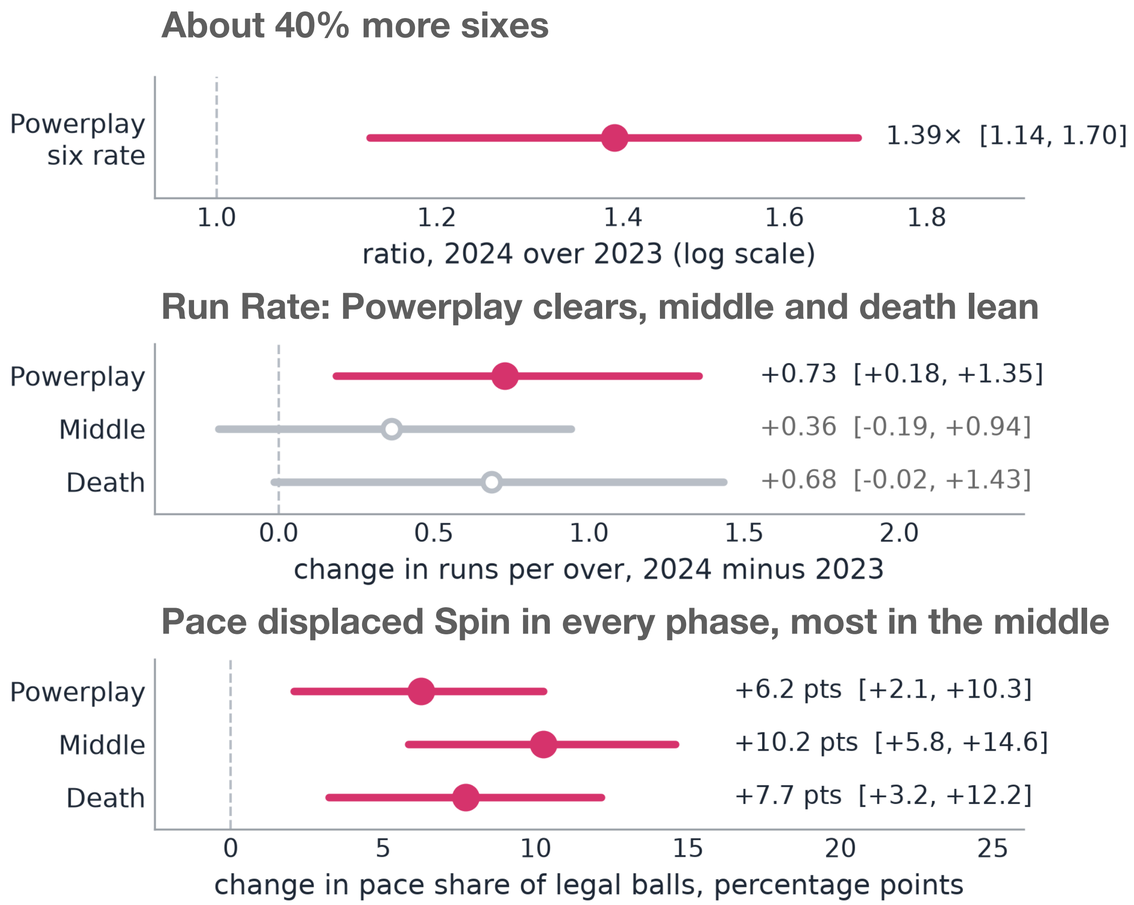
## Analytical Methods

*Matches as the unit, intervals whose coverage is checked, exact decompositions, and models validated on planted answers*

- Match cluster bootstrap for every season contrast on shared draws; bootstrap-t and BCa intervals, coverage verified by simulation at 70 and 14 clusters.
- Equivalence verdicts with pre-stated smallest effects, so a null is evidence of absence or is reported as unresolved with its minimum detectable effect.
- Exact decompositions (shift-share, Shapley) to say where a change came from: same or different batters, mix or menu, sixes or dots, selection or benching.
- Hierarchical models with two priors for between-team spread; a leakage audited situational expected-wicket model and a win-probability ledger to price the headline; a slope-change detector with a measured resolution for the phase grid.
- For totals, the death overs and phase boundaries, only populations a Chase cannot truncate (first innings, full allowance) used.

Github Repository

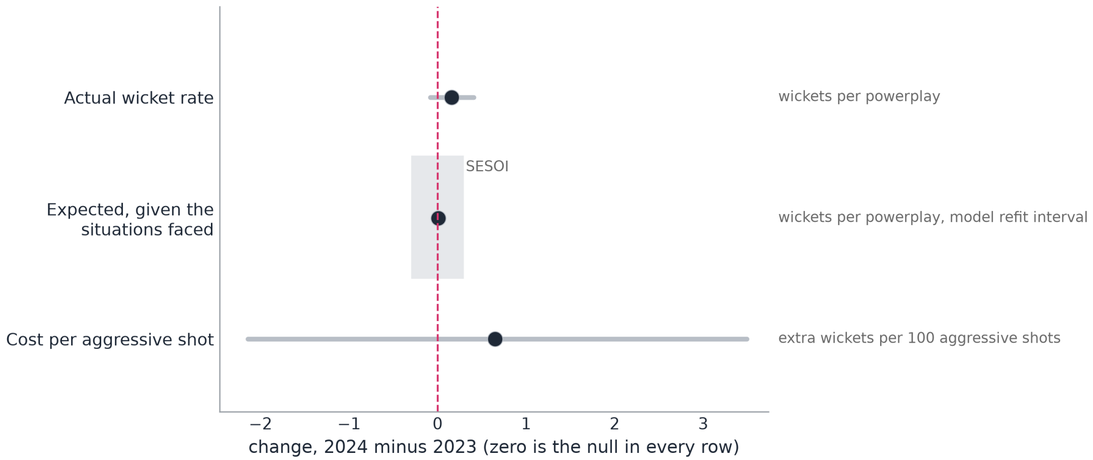
# 2024 scored faster runs in every phase; the extra runs were sixes



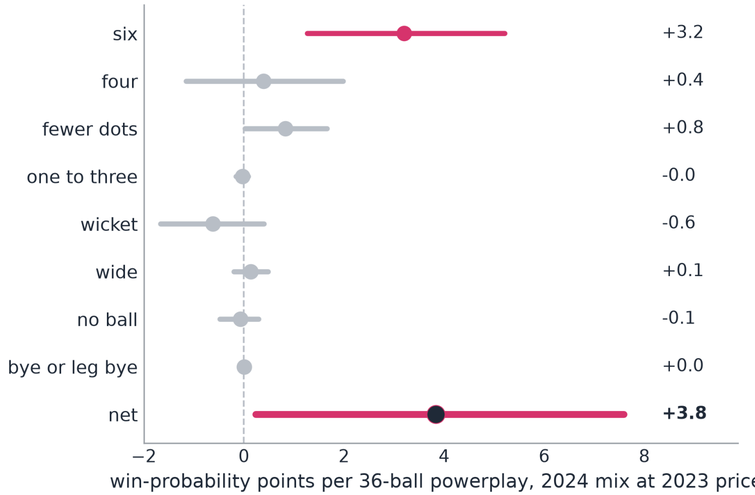



# Extra risk was priced and it paid about 4 win probability points per Powerplay

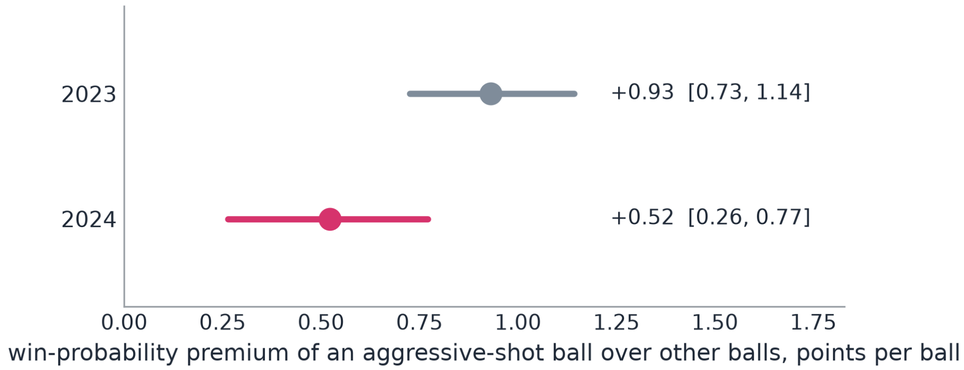
## No wicket cost at Powerplay



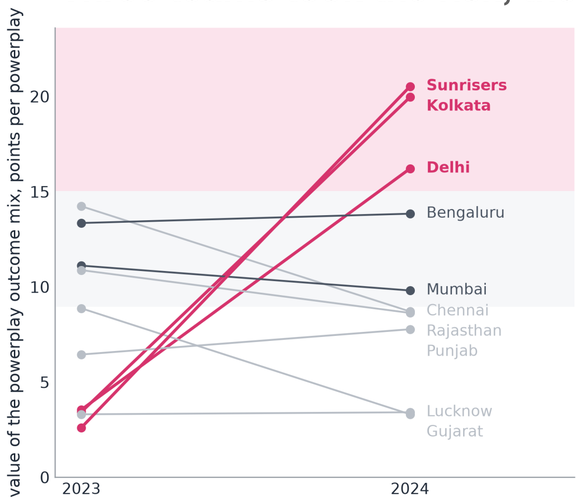

## 2024 mix was worth +3.8 wprob




## Diminishing returns for aggressive shots in 2024

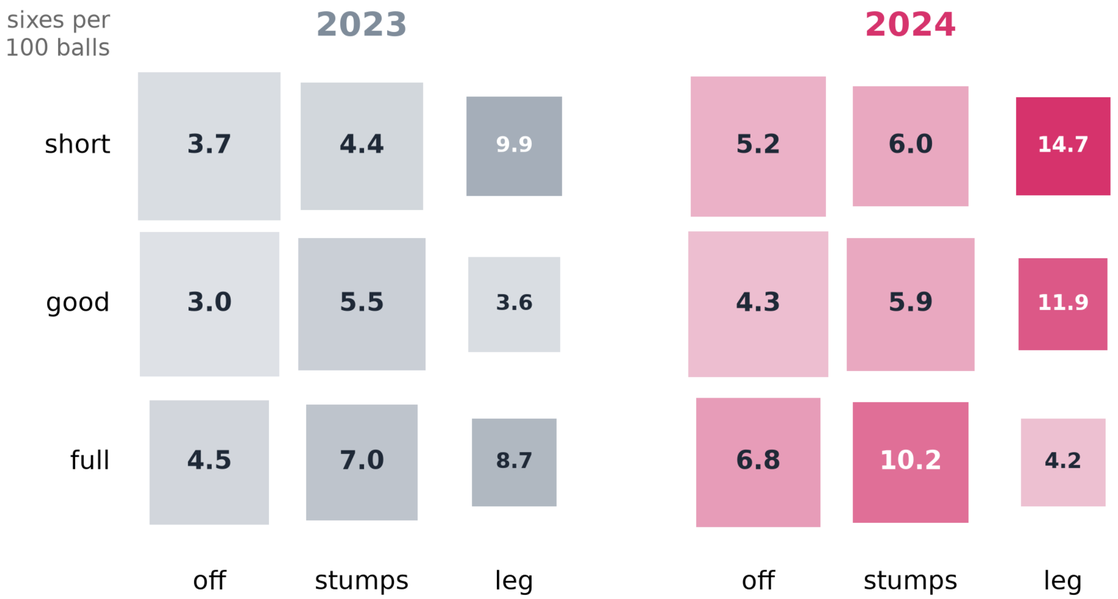



## Three teams took the risk; the order reversed



# No Line and Length Answer: Every delivery was hit harder, Spin was benched

## Bowlers bowled fuller and faster, but it made little difference




## Pace was the target; slower ball held but was cut




## Spin was deployed less and benched more



# Same openers, bought hitters and Chases that went hard from ball one

## Half the Powerplay rise was three openers; most of the other half was bought

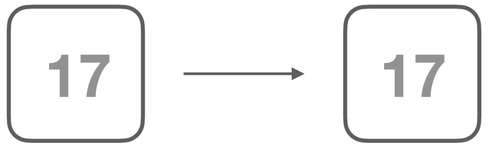



## Chasing sides went harder at every target, decided this early




## Impact Substitute: Warm-up unchanged, twelfth batter insurance is new

Bowling-first sides, during fielding innings




Batting-first sides, during their own batting innings



# Strengths and Weaknesses of Chosen Methodology

## Strengths

- The unit of inference matches the structure of the data, and the intervals' coverage was checked rather than assumed
- Every claim carries a stated status and every null carries a stated smallest effect and minimum detectable effect, so absence of evidence and evidence of absence are kept apart
- Populations are chosen to avoid the chase censoring that would otherwise bias totals, death overs and profiles, and the exclusions are counted.
- Decompositions are exact and verified on every resample.
- Where raw data and EDA suggested a narrative (bowlers bowling into the surge), the exact statistical decomposition was allowed to overrule it
- Model-based methods were validated on planted answers before they touched real data, and one method's resolution was measured and reported as a limit
- Everything reproduces from a raw file, a config and a seed, with provenance on every output

## Weaknesses

- Two seasons provide a step, not a trend, and no causal claim is possible
- The team-level work is underpowered by construction (fourteen matches per team), and its established results rest partly on a hand-tagged column
- Several results depend on the provided win-probability and expected-wicket ledgers being league-average and state-average accountings; a six at 0 for 2 is priced like one at 60 for 0.
- Intervals are resampling stability, not replication
- The break detector resolves two overs, not one, so a one-over boundary move would be invisible to it
- Uncensored populations are cleaner but is only half the sample, resulting in reduced statistical power.

# Interpretation and Practical Significance

## Cricket Interpretation

- Powerplay was treated as a window to clear the infield rather than find gaps. Sixes rose by 40%, boundaries made the extra runs in 2024 in every phase.
- Situations teams batted in were not safer and actual wicket rate did not move much in 2024. Attacking shots produced a mix that was worth about 4 win-probability percentage points but at the same wicket risk.
- Half the Powerplay rise was three continuing openers and the other half was recruiting six hitters for the top. Lower order change in six hitting mostly made by a change in who was sent in to bat.
- Every class of delivery was hit for more sixes in 2024. Over 140kmph balls were most hit; slower and good-length-on-stump balls held but slow balls were bowled about a third less.
- Chasing sides attacked in the Powerplay harder than in target setting in both seasons, and went harder at every target in 2024
- Batting first sides added a batter as impact substitute mid-innings who mostly did not bat and did not score quickly when he did. Impact substitute was used mostly as an insurance.
- League pulled apart into tiers in the Powerplay, teams chose to be very aggressive (SRH, DC, KKR) or conservative (LSG, GT). Middle and Death overs were navigated similarly by all teams.

## Practical Relevance

- Sides without six hitting openers are giving up on the phase where risk is cheap. Franchises should plan for Powerplay hitting as a specialised role.
- Since all line-and-length combinations were conceding more in 2024, and slower balls held + took more wickets, bowlers who can take the pace off need to be deployed in the Powerplay
- Teams kept paying for spinners and benched them or were selected and deployed less in 2024. Franchises should decide before auction on whether to carry fewer spinners and spend elsewhere or to have a focused selection and deployment strategy for spinners.
- Each aggressive shot bought about half as much win-probability in 2024 as in 2023 as the baseline for league rose. The return came from volume and will do so again in 2025 but franchise will need to expect even fewer win-probability points per aggressive shot in the future.
- What a team should not conclude: that the acceleration was bookended, that 220 became routine (one in ten to one in five on first innings), that the death overs converged, that the phase boundaries moved, or that bowlers bowled into the surge. The fixed phase plan is still the right frame.

# References and Code Repository

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- https://github.com/madkad-ds/dpl_2026_s1/ (Requirements and Reproducibility instructions provided in README)